

Pattern Dry Run Test Section A: Predict the Output (Code only, no answers)

Q1.

```
n = 5
for i in range(1, n+1):
    for j in range(1, i+1):
        print((i+j)%2, end=" ")
    print()
```

Q2.

```
n = 5
for i in range(n):
    for j in range(n):
        if i == j or i+j == n-1:
            print(j, end="")
        else:
            print(" ", end="")
    print()
```

Q3.

```
n = 5
for i in range(n):
    for j in range(2*n-1):
        if j >= n-i-1 and j <= n+i-1:
            print(i+1, end="")
        else:
            print(" ", end="")
    print()
```

Q4.

```
n = 5
for i in range(n):
    for j in range(n):
        if i == j or i+j == n-1 or i == 0 or i == n-1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

Q5.

```
n = 4
k = 1
for i in range(n):
    if i % 2 == 0:
        for j in range(n):
            print(k, end=" ")
            k += 1
    else:
        temp = k + n - 1
        for j in range(n):
            print(temp, end=" ")
            temp -= 1
            k += 1
print()
```

Q6.

n = 5

```
for i in range(1, n+1):  
    for j in range(n-i):  
        print(" ", end="")  
    for j in range(i, 0, -1):  
        print(j, end="")  
    for j in range(2, i+1):  
        print(j, end="")  
    print()
```

LLM Foundations Assessment

1. What is the primary focus of Generative AI?

- A. Storing databases
- ☒ B. Creating new outputs
- C. Increasing internet speed
- D. Managing operating systems

2. Which of the following best describes an LLM?

- A. A hardware acceleration chip
- B. A search engine database
- ☒ C. An AI model designed to understand and generate language
- D. A cybersecurity firewall

3. According to the session, LLMs mainly generate responses by:

- A. Thinking exactly like humans
- ☒ B. Predicting the most likely next words
- C. Searching Google every time
- D. Reading human emotions directly

4. Which factor was NOT mentioned as something that changes between AI models?

- A. Creativity
- B. Sentence structure
- ☒ C. Battery optimization
- D. Logic and reasoning

5. Why can two AI tools give completely different answers to the same prompt?

- ☒ A. Every LLM is built differently
- B. Every user has different internet speed
- C. AI tools only work randomly
- D. The keyboard affects the output

6. Which option BEST improves AI output quality?

- A. Asking very short random questions
- ☒ B. Providing clear context and goals
- C. Using capital letters everywhere
- D. Repeating the same prompt multiple times

7. What changed in the startup idea example after additional details were added?

- A. The internet speed increased
- B. The AI model changed automatically
- ☒ C. The context became clearer
- D. The AI stopped generating outputs

8. Which of the following is an example of context?

- A. "Suggest a startup idea."
- ☒ B. "I am a B.Tech student interested in AI and low-investment ideas."
- C. "Tell me something."
- D. "Write fast."

9. Which statement about LLMs is MOST accurate according to the session?

- A. LLMs always provide factually correct answers
- B. LLMs think like humans
- ☒ C. LLMs can generate confident but incorrect information
- D. LLMs only work for coding tasks

10. What is an AI hallucination?

- A. AI shutting down unexpectedly
- ☒ B. AI generating false or assumed information
- C. AI learning emotions permanently
- D. AI refusing to answer prompts

11. Why is verification important while using AI?

- A. AI may generate fake facts confidently
- ☒ B. AI only works with verification
- C. AI cannot process language without verification
- D. Verification improves Wi-Fi speed

12. A student asks AI:

"I missed one semester. Can I still get a software job?"

- ☒ A. AI may assume missing details incorrectly
- B. AI cannot understand career questions
- C. AI only responds to technical prompts
- D. AI will automatically reject the question

13. What is a knowledge cutoff?

- A. AI memory getting deleted
- ☒ B. The point until which the model has learned information
- C. The maximum number of prompts allowed
- D. A limit on internet access speed

14. Information after an AI model's knowledge cutoff may become:

- A. More accurate automatically
- B. Completely encrypted
- ☒ C. Unknown, partially known, or inaccurate
- D. Deleted permanently from the internet

15. Which approach helps AI perform better?

- A. Vague prompts without details
- ☒ B. Progressive conversation with follow-up questions
- C. Asking unrelated questions repeatedly
- D. Avoiding context completely

16. Which of the following was directly mentioned as training data for LLMs?

(Select all that apply)

- A. Books
- B. Research papers
- C. Public internet text
- ☒ D. All the above

17. Which question is MOST useful to identify whether an AI tool is using live information?

- A. "What color is AI?"
- B. "Who invented Wi-Fi?"
- ☒ C. "Are your answers based on pretrained data or live search?"
- D. "Is the output right?"

18. What does MCP stand for?

- A. Machine Context Processing
- B. Multi Communication Platform
- ☒ C. Model Context Protocol
- D. Managed Computing Process

19. Why is MCP important in modern AI systems?

- ☒ A. It helps AI systems access better context and external tools
- B. It increases laptop RAM automatically
- C. It converts AI into hardware devices
- D. It replaces all prompts completely

20. Which student is MOST likely to get better AI outputs?

- A. A student using one-line vague prompts
- ☒ B. A student giving detailed context, goals, and follow-up questions
- C. A student copying random prompts without understanding
- D. A student asking unrelated questions continuously